

Data Analytics Immersive

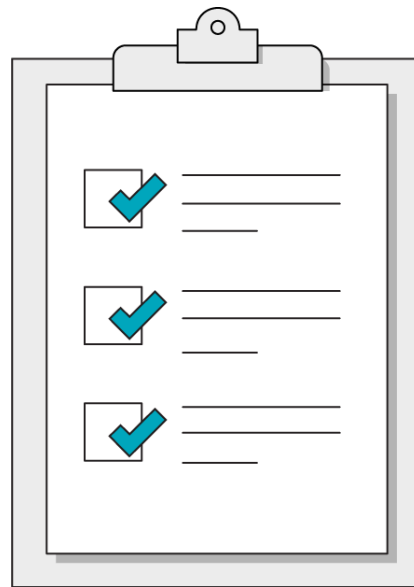
Introduction to Modeling and Machine Learning

Unit: Statistics and mathematics for data analysts

Learning Objectives

In this lesson, we'll:

- **Learn** what machine learning is (and what it is not), and how it differs from statistical modeling.
- **Understand** the difference between supervised, unsupervised and reinforcement learning.
- **Discuss** the application of machine learning models and their real world relevance.



Any questions?

Agenda



Key Terminology



What is Machine Learning?



Supervised and Unsupervised Learning

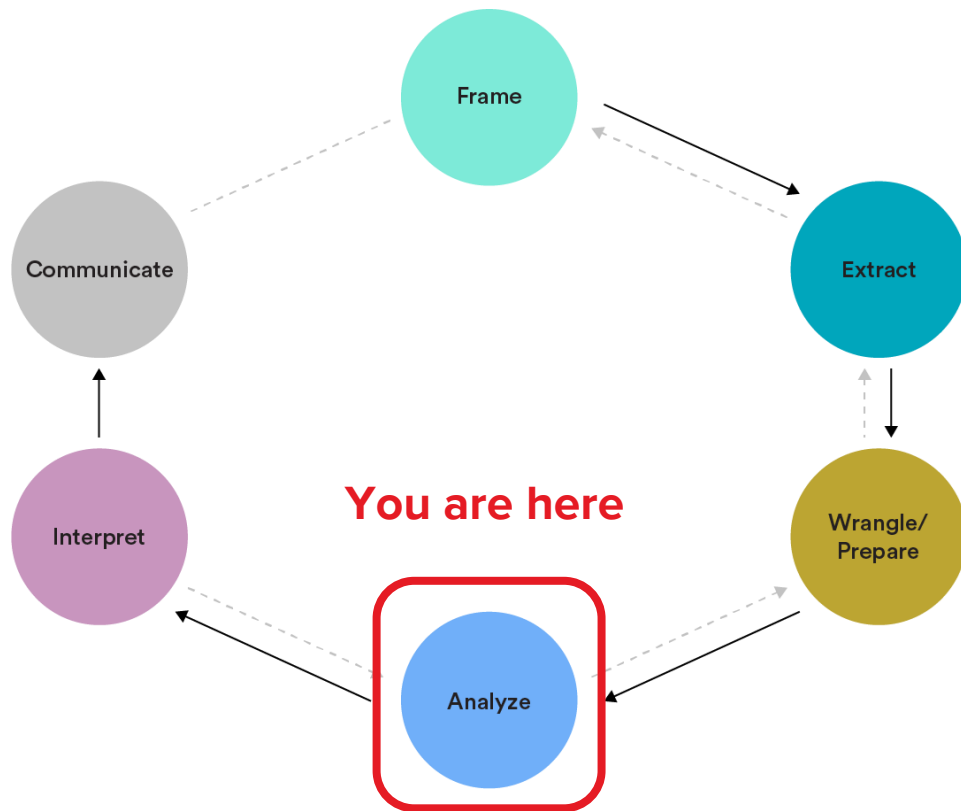


Reinforcement Learning



Case Studies of Using Machine Learning

Where Are We in the DA Workflow?



Frame: Develop hypothesis-driven questions for your analysis.

Extract: Select and import relevant data.

Wrangle/Prepare: Clean and prepare relevant data.

Analyze: Structure, comprehend, model and visualize data.

Interpret: Leverage your analysis to make decisions and recommendations.

Communicate: Present data-driven findings and insights in a compelling manner.

Introduction to Modeling and Machine Learning

Key Terminology



Terminology

Variable

Measurements of
a single quantity,
taken from many
subjects, e.g.
company share
price measured
across 1,000
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Dependent variable

Also known as:
target/response/label/output

A variable we want to predict, e.g.
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Also known as:
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Independent variable

Also known as:
**feature/predictor/
covariate/input)**

A variable we're using to predict something, e.g. company profits

Terminology

Variable

Measurements of a single quantity, taken from many subjects, e.g. company share price measured across 1,000 companies.

Dependent variable

Also known as:
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A variable we want to predict, e.g. company share price.

Independent variable

Also known as:
feature/predictor/covariate/input)

A variable we're using to predict something, e.g. company profits.

Model

A mathematical relationship between two variables. Models can be very simple, or very complex.



Discussion: Terminology

5 minutes



Stamp with **Annotate** option

Put a ♥ stamp next to the **independent variable**, and a ➡ stamp next to the **dependent variable** for each of these cases:

- Modeling the effect of **coffee consumption** on a **student's final grade**.
- Predicting a **company's share price** using their **profits**.
- Predicting a **company's gender pay gap** using the **proportion of women on the company's board** and the **company sector**.

Terminology

Variables can be continuous or categorical.

Continuous Variables

Numerical variables, with an infinite number of values between any two values. For example, there are an infinite number of values between a height of 2 meters and 2.1 meters, so height is a continuous variable.

Categorical Variables

Variables with a finite number of categories or groups. For example, t-shirt size can be small, medium or large.



Stamp with **Annotate** option

Are these variables **categorical** or **continuous**?

Put a stamp in each row to cast your vote.

Variable	Categorical	Continuous
Height		
Eye color		
Weight		
Month		



Discussion:

Categorical or Continuous?



Stamp with **Annotate** option

Are these variables **categorical** or **continuous**?

Put a stamp in each row to cast your vote.

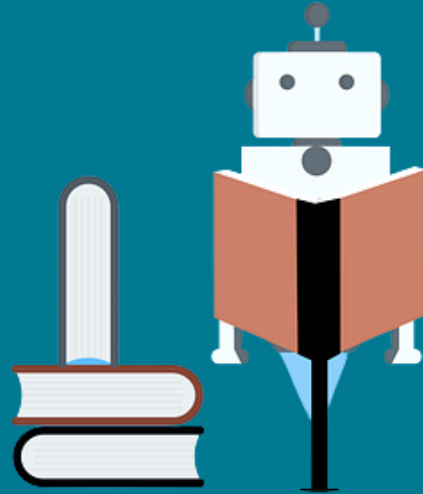
Variable	Categorical	Continuous
Height		
Eye color		
Weight		
Month		

Introduction to Modeling and Machine Learning

What Is Machine Learning?



Let's start to unpack what machine learning is, how it works, and what it's used for.





Solo Exercise:

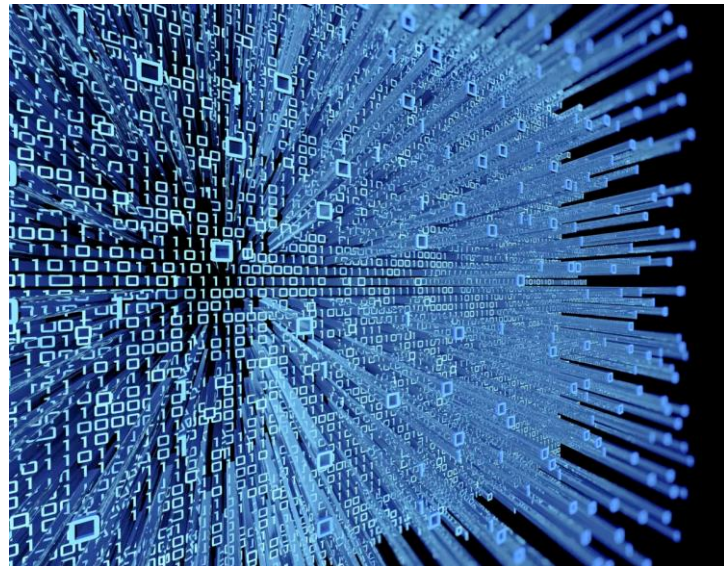
What Is Machine Learning?

5 minutes

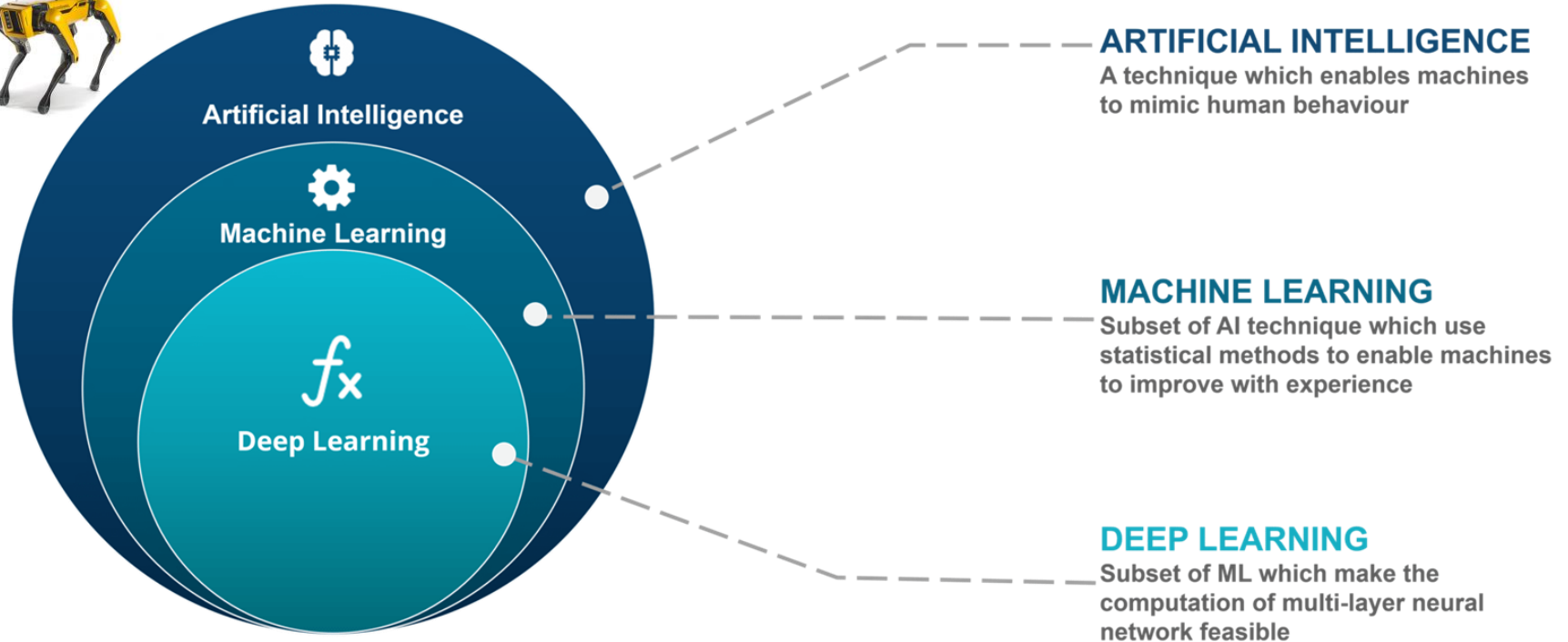


 **In Slack**, post any preconceptions you have about:

- What machine learning is.
- What it's used for.
- How it works.
- Why there's a lot of hype about it.
- How it differs from artificial intelligence, big data and deep learning.



A Few Definitions





Group Exercise: Chair or Not?

5 minutes



Imagine you're teaching a small child to recognize pictures of chairs.

Think of some **rules** you might teach the child to help them distinguish **chairs** from **not-chairs** (e.g. if it's made of wood then it's a chair).

 Post your set of rules in **Slack** when you're done.



Chair or Not?

All the pictures on the right are of chairs.

Would your rules correctly identify all of them?

Rule-based systems aren't very robust.



Chair or Not?

In real life, how might a child actually learn to recognize chairs?

You'd probably just show them lots of examples of chairs and say "That's a chair" each time.

They would eventually **learn their own** set of rules for identifying chairs.

When they saw a new style of chair they'd never seen before, they would use these **learned** rules to correctly identify it.

All you need to do is give them lots of examples to learn from!



These
are all
chairs.



That's
a chair!



Logical Reasoning vs. Machine Learning

Two different approaches to the problem, “How do we map an input to an output?”

(E.g., how do we map a house’s size to its sale price?

how do we map a picture to whether it’s a chair or not?)

Logical reasoning

Use expert knowledge to write a set of rules that convert inputs to outputs.

Sounds intuitive but isn’t very robust (as we’ve seen with the chair example).

Machine learning

Show the learner lots of examples of inputs and corresponding outputs.

The learner finds the relationship between the inputs and the output.

When shown a new input, it’s able to use this learned relationship to predict the output.

Machine Learning

We've seen that if we're trying to teach someone to map an **input** to an **output** then explicitly giving them a set of rules to do this isn't very robust or efficient.

In the chair example:

The **input** is the picture.

The **output** is whether it's a chair or not.

In the house example:

The **input** is information about a house including its size, location, etc..

The **output** is how much it will sell for.



It's much better to **train** people with lots of **examples** and let them figure out the relationship between the input and the output themselves.

It's exactly the same when we're teaching computers to do things. This is what machine learning is all about!

Can you think of other words to describe **input**, **output** and **relationship**?



Machine Learning

By learning the relationship between independent and dependent variables using **training examples** or **training data** a computer program can:

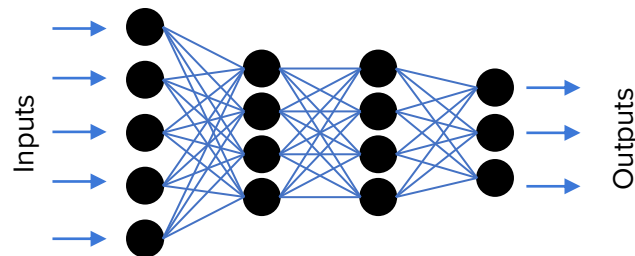
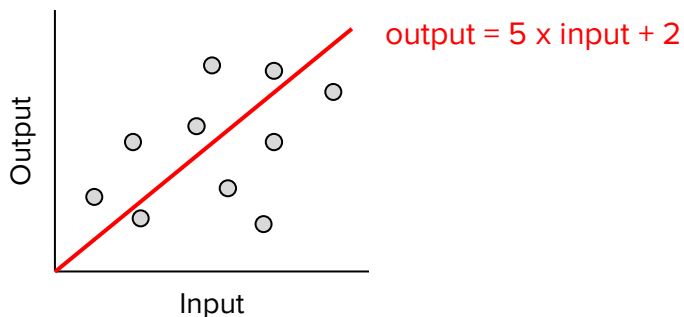
- Make predictions about the value (e.g. sale price) or class (e.g. chair or not-chair) of data points it hasn't seen before.
- Automate some manual tasks.
- Find groupings and patterns in data.

The **learning** in machine learning just means the computer is using training data to find a **mathematical model or relationship** that links the independent and dependent variables.

Machine Learning

This model might be a very simple straight-line model (this is **linear regression**) or a very complex model (like a **neural network**).

Choosing the right sort of model for your data is an important decision (fancier models are **not** always better).



The Three Common Types of Machine Learning

Method 1:



Give them **labeled** pictures of chairs.

After studying those pictures, they should be able to look at new pictures and predict if it's a chair or not.

Supervised Learning

Method 2:

Give them **unlabeled** pictures of chairs and other items.

They should be able to sort the pictures into groups of chairs or not-chairs based on similar characteristics.

Unsupervised Learning

Method 3:

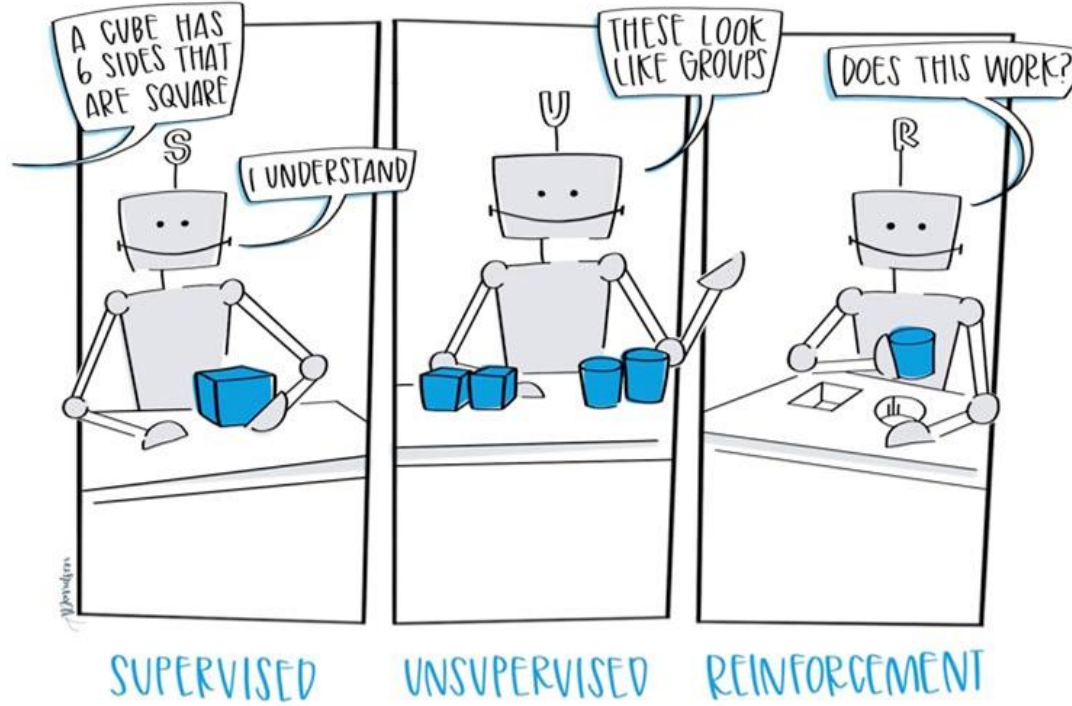
Reward them.



Have them guess which images are chairs. Reward them when they get it right, and correct them when they get it wrong as they continue guessing.

Reinforcement Learning

MACHINE LEARNING



Introduction to Modeling and Machine Learning

Supervised and Unsupervised Learning



Supervised vs. Unsupervised Machine Learning

Supervised

- Using independent variables to predict a dependent variable.

Vs.

Unsupervised

- Finding groups and patterns in data.



Supervised Machine Learning

Finding a model that predicts **y (a response)** using values of **x (features)**.

We use a **training data set** which contains many corresponding values of **y** and **x** to find the mathematical relationship between our variables.

This is called **training** or **fitting** a model.

After our model has been trained, it can predict values of **y** when it's given new values of **x**.

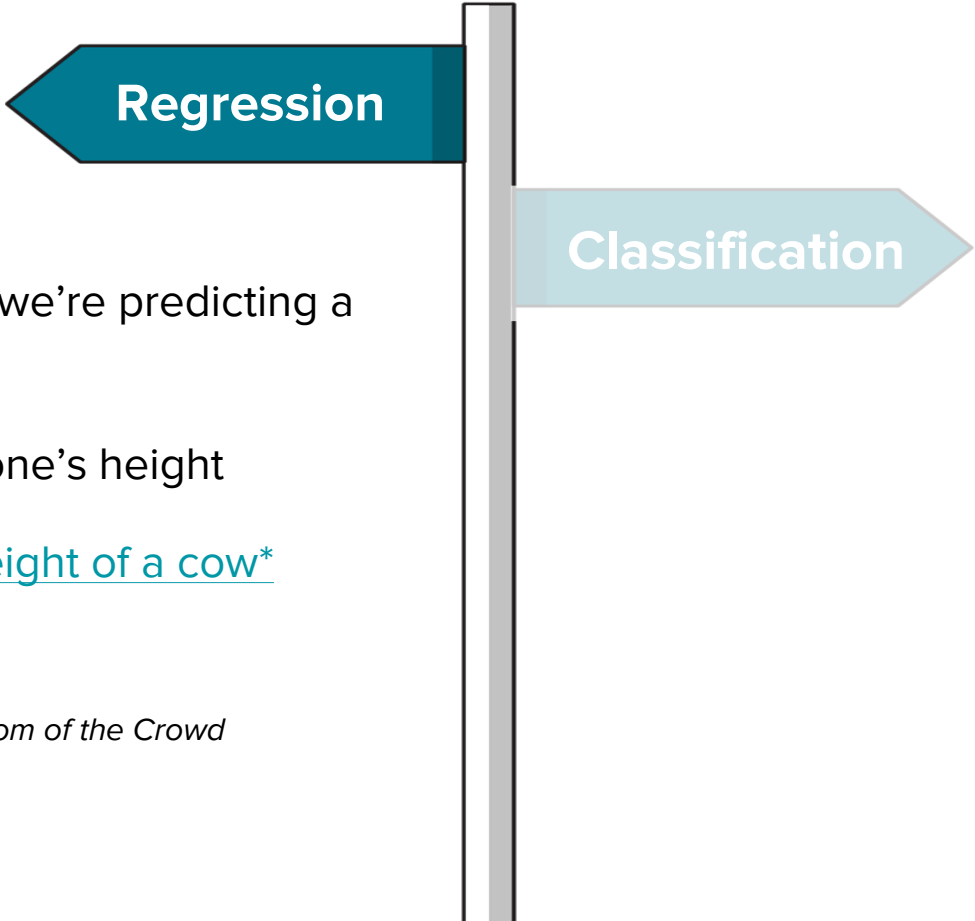
Supervised Machine Learning

Problem: Predict house price using internal area.

This is a **supervised learning** task because it fits the pattern of 'predict y using x '.



Regression vs. Classification



Regression

Classification

In **regression** tasks, we're predicting a **continuous** output:

- Predicting someone's height
- Predicting the weight of a cow*

** Great podcast on the Wisdom of the Crowd*

Regression vs. Classification

Regression

In **regression** tasks, we're predicting a **continuous** output:

- Predicting someone's height
- Predicting the weight of a cow*

** Great podcast on the Wisdom of the Crowd*

Classification

In **classification** tasks, we're predicting a **categorical** output:

- Predicting the winning party in an election
- Predicting whether a customer will buy a product or not



Solo Exercise:

Classification or Regression?

5 minutes



For each of these machine learning problems, put a ❤️ stamp next to the **feature(s)** and a ➡️ next to the **response**, then decide whether it's a classification or regression task.

Problem	Classification	Regression
Predicting daily rainfall (mm) using temperature, air humidity and wind speed.		
Predicting a house sale price using its location and number of bedrooms.		
Predicting a hair color using a person's eye color and age.		
Forecasting tech sales on Black Friday.		
Predicting power plant failures.		



Solo Exercise:

Classification or Regression? Answers

5 minutes



For each of these machine learning problems, put a ❤️ stamp next to the **feature(s)** and a ➡️ next to the **response**, then decide whether it's a classification or regression task.

Problem	Classification	Regression
Predicting daily rainfall (mm) using temperature , air humidity and wind speed. ➡️ ❤️ ❤️ ❤️		★
Predicting a house sale price using its location and number of bedrooms.		★
Predicting a hair color using a person's eye color and age.	★	
Forecasting tech sales on Black Friday.		★
Predicting power plant failures.	★	



Unsupervised Machine Learning

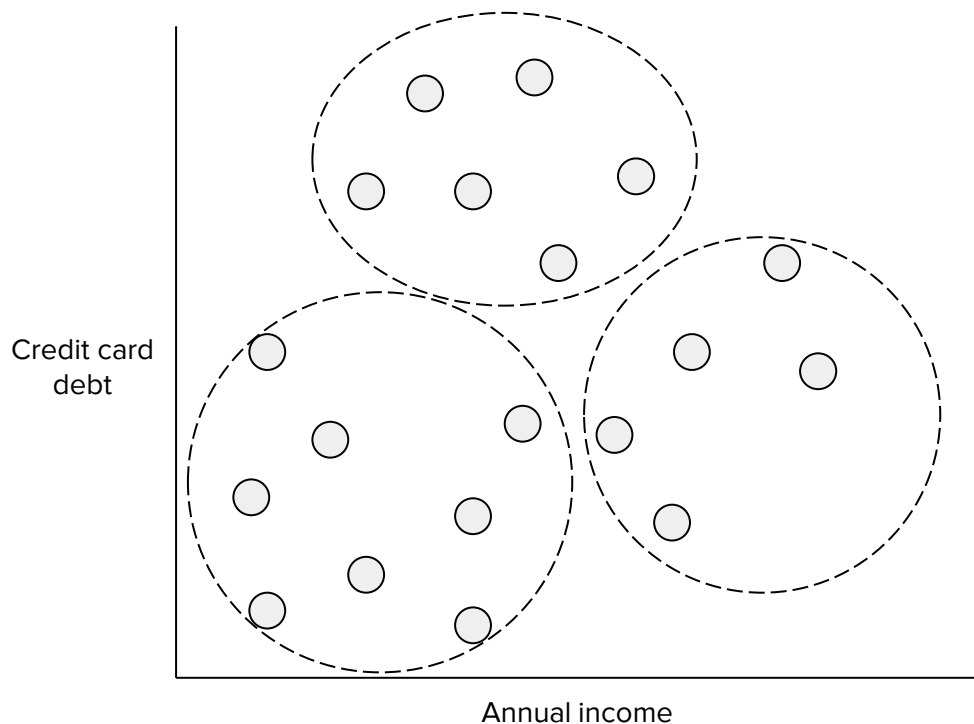
This is used for finding **clusters**, **groups** and **patterns** in data sets.

It's also used for **dimensionality reduction**, or reducing the number of variables in a data set without losing useful information.

We're **not** explicitly predicting something.

The results of unsupervised learning often involve a lot of **domain expertise** and **human interpretation**.

Unsupervised Learning



Problem: Find meaningful clusters in customer data.

Use an **unsupervised** clustering method to find groupings in the data.

In this case, the algorithm has found three **clusters**.

Human interpretation is needed to make sense of what these clusters mean.



Discussion:

Steps for Clustering

1 minute



Imagine that we had a bunch of coins we wanted to automatically split into groups.

How would an unsupervised learning technique cluster the coins?

What would be the end result?





Steps for Clustering

An unsupervised learning technique would:

1. Cluster the coins based on **similarity**.
This is not pre-defined but determined by the algorithm, and it could sort them through the size, the material, or the language on the coins, or other factors.
1. Inspect the grouping that the algorithm found.

This would put the coins into sets of related groups that were hopefully easy to interpret (e.g. size and material differs across groups).





Unsupervised Learning at Netflix

Has Netflix ever recommended a genre of content to you that's weirdly, eerily, and extremely specific and detailed? That's powered by unsupervised learning.



Data used: Content data (e.g., genre, categories, actors), your viewing history, your viewing habits (e.g., when, how long, devices), where you live, how you rate shows, similar viewers' preferences, and more.

Output: Netflix's 167 million users are clustered into over 2,000 "taste groups" of viewers who enjoy and are likely to watch similar content.

Result: Taste groups are behind Netflix's recommendation system, and 80% of shows watched on the platform are found via this system.

Introduction to Modeling and Machine Learning

Reinforcement Learning



Reinforcement Learning

The main difference between **supervised learning** and **reinforcement learning** is how a model is created.

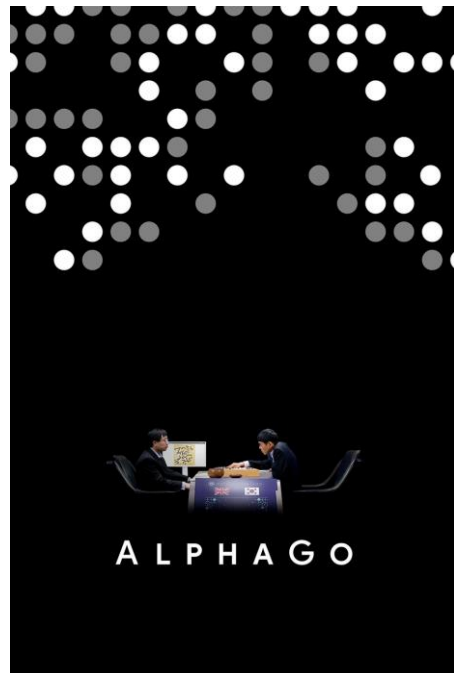
Rather than learning from a training data set, in reinforcement learning an objective is set and the algorithm seeks to reach the objective by altering parameters within the model. It receives feedback (the **reward**) on whether its decisions were good or bad. It then adjusts parameters accordingly.



Reinforcement Learning

Reinforcement learning is often used to play games: AlphaGo by DeepMind beat the world champion of Go in 2016 using reinforcement learning, something which had seemed impossible even months before. Its next iteration, AlphaGo Zero, beat AlphaGo 100 to zero by playing games against itself.

It is beyond the scope of this course to go into reinforcement learning, but if you want more information on it then [this](#) is a good place to start.





Real Cases:

Cooling Data Centers with Reinforcement Learning

The needs and requirements of Google's data centers are too complex to manage using pre-canned approaches or human intuition.

Instead, they use reinforcement learning to manage the environment more efficiently.

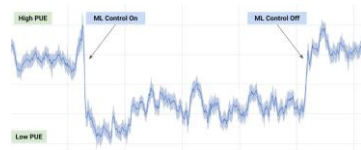




Data used: Historical data (e.g. temperatures, power, pump speeds, setpoints) updated with sensor readings every five minutes. This data is fed into deep neural networks which predict how different combinations of potential actions will affect future energy consumption.

Output: Operational settings are refined to maintain temperature and minimize energy usage. The model is allowed to control the system settings, responding to inputs and learning from results.

Result: **30–40%** energy savings per data center.



Introduction to Modeling and Machine Learning

Case Studies of Using Machine Learning





Discussion: Training Data

5 minutes



Supervised machine learning relies on having a large set of **training data** that a computer can use to find the relationship between an input and an output.

Each data point also needs to be labeled.

For example, to teach a computer to detect photos of different models of cars our training data would consist of thousands of images of cars together with the model.



What are some potential problems with this?

Predictive Models

Most predictive models are either **classifiers** where the target variable is discrete, or **regression models** if the target variable is continuous.

Powerful models don't always have to be very complicated; in fact some of the most widely used models are very simple.

Let's explore one of them.



Real Cases:

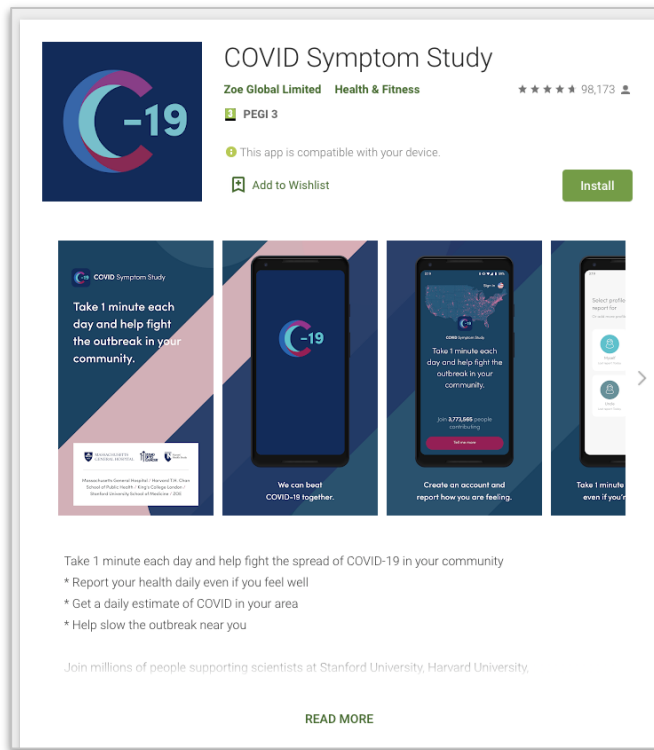
Covid Symptom Tracker

In 2020, a group at King's College London released a Covid symptom tracker app.

Every day, millions of people in the UK enter their symptoms (or lack of symptoms) into the app, together with information about their age, gender and other characteristics.

<https://covid.joinzoe.com/>

The researchers used this data to build a **model that predicts** the probability that a person has coronavirus, using information about their gender, age and symptoms.





Let's take a look at the detail behind this press release from King's College London to understand exactly how the model works and how accurate it is.

<https://covid.joinzoe.com/post/detecting-covid-without-a-test>

Detecting COVID-19 using AI

May 15, 2020

New AI diagnostic can predict COVID-19 likelihood without a test

London and Boston – Researchers at health science company ZOE in collaboration with Massachusetts General Hospital, King's College London, and the University of Nottingham have developed an artificial intelligence diagnostic that can predict whether someone is likely to have COVID-19 based on their symptoms. The initial phase of the research is now published in the leading medical journal, Nature Medicine.



Solo Exercise:

Coronavirus Infection Model

10 minutes



Have a read through the Nature paper published by a group from King's College in London:

<https://www.nature.com/articles/s41591-020-0916-2>

- Can you find the equation that describes the model?
- What was the training data set? How was it compiled?
- Is the equation easy to understand?
- Where does 'AI' come into the process?

nature medicine

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Brief Communication | [Published: 11 May 2020](#)

Real-time tracking of self-reported symptoms to predict potential COVID-19

[Cristina Menni](#) , [Ana M. Valdes](#), [Maxim B. Freidin](#), [Carole H. Sudre](#), [Long H. Nguyen](#), [David A. Drew](#), [Sajaysurya Ganesh](#), [Thomas Varsavsky](#), [M. Jorge Cardoso](#), [Julia S. El-Sayed Moustafa](#), [Alessia Visconti](#), [Pirro Hysi](#), [Ruth C. E. Bowyer](#), [Massimo Mangino](#), [Mario Falchi](#), [Jonathan Wolf](#), [Sebastien Ourselin](#), [Andrew T. Chan](#), [Claire J. Steves](#) & [Tim D. Spector](#) 

[Nature Medicine](#) **26**, 1037–1040 (2020) | [Cite this article](#)

241k Accesses | **576** Citations | **2931** Altmetric | [Metrics](#)

Abstract

A total of 2,618,862 participants reported their potential symptoms of COVID-19 on a smartphone-based app. Among the 18,401 who had undergone a SARS-CoV-2 test, the proportion of participants who reported loss of smell and taste was higher in those with a positive test result (4,668 of 7,178 individuals; 65.03%) than in those with a negative test result (2,436 of 11,223 participants; 21.71%) (odds ratio = 6.74; 95% confidence interval = 6.31–7.21). A



Solo Exercise:

Coronavirus Infection Model: Answers

This is the model! It's very simple — just a linear combination of inputs like age, gender, and symptoms — but still useful and powerful in predicting coronavirus infection.

To calculate the probability that the person has coronavirus, compute $e^{\text{prediction}} / (1 + e^{\text{prediction}})$

Sex is 1 if the person is male, 0 if the person is female

$$\begin{aligned} \text{Prediction} &= -1.32 - (0.01 \times \text{age}) \\ &+ (0.44 \times \text{sex}) + (1.75 \times \text{loss of smell and taste}) \\ &+ (0.31 \times \text{severe or significant persistent cough}) \\ &+ (0.49 \times \text{severe fatigue}) + (0.39 \times \text{skipped meals}) \end{aligned}$$

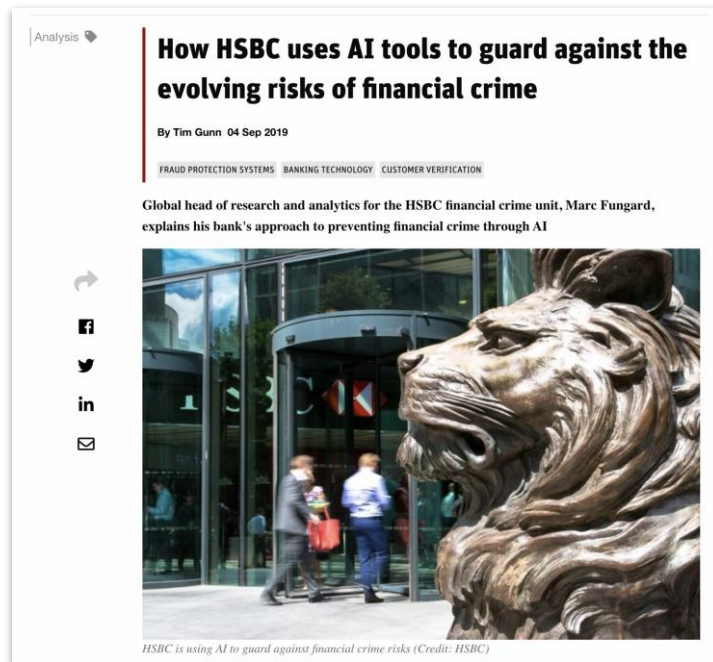
Symptoms take the value 1 if a person has that symptom, and 0 otherwise



How does a single human sort through petabytes of data to find money laundering, international fraud, and terrorist activity (one petabyte is approx. 225,000 copies of “Titanic”)?

Automate the identification of key signals from payment volume, recipient, amount, account creation, and more.

<https://www.nsbanking.com/analysis/hsbc-financial-crime-ai/>

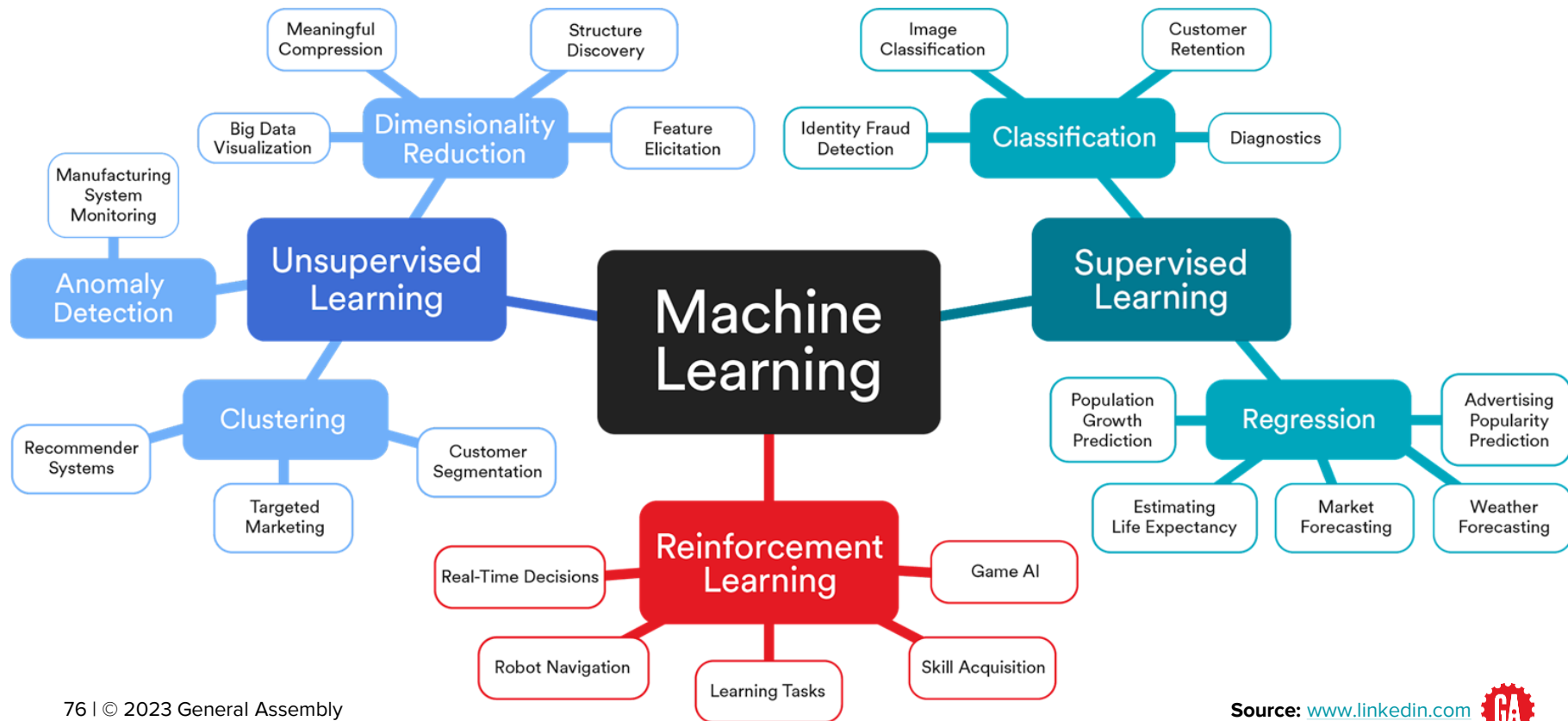


Introduction to Modeling and Machine Learning

Wrapping Up



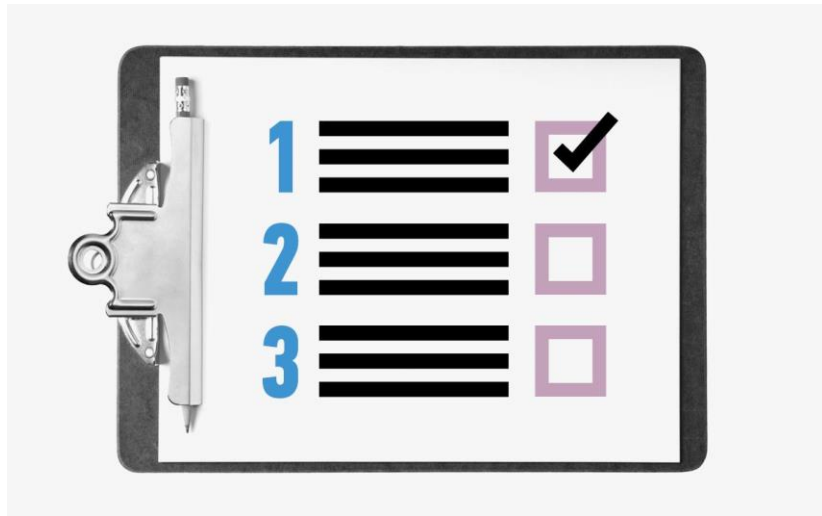
Putting it all together (in a bit more detail)



Recap

Today, we:

- Learned what machine learning is (and what it is not), and how it differs from statistical modeling.
- Understood the difference between supervised, unsupervised and reinforcement learning.
- Discussed the application of machine learning models and their real world relevance.



Q&A

